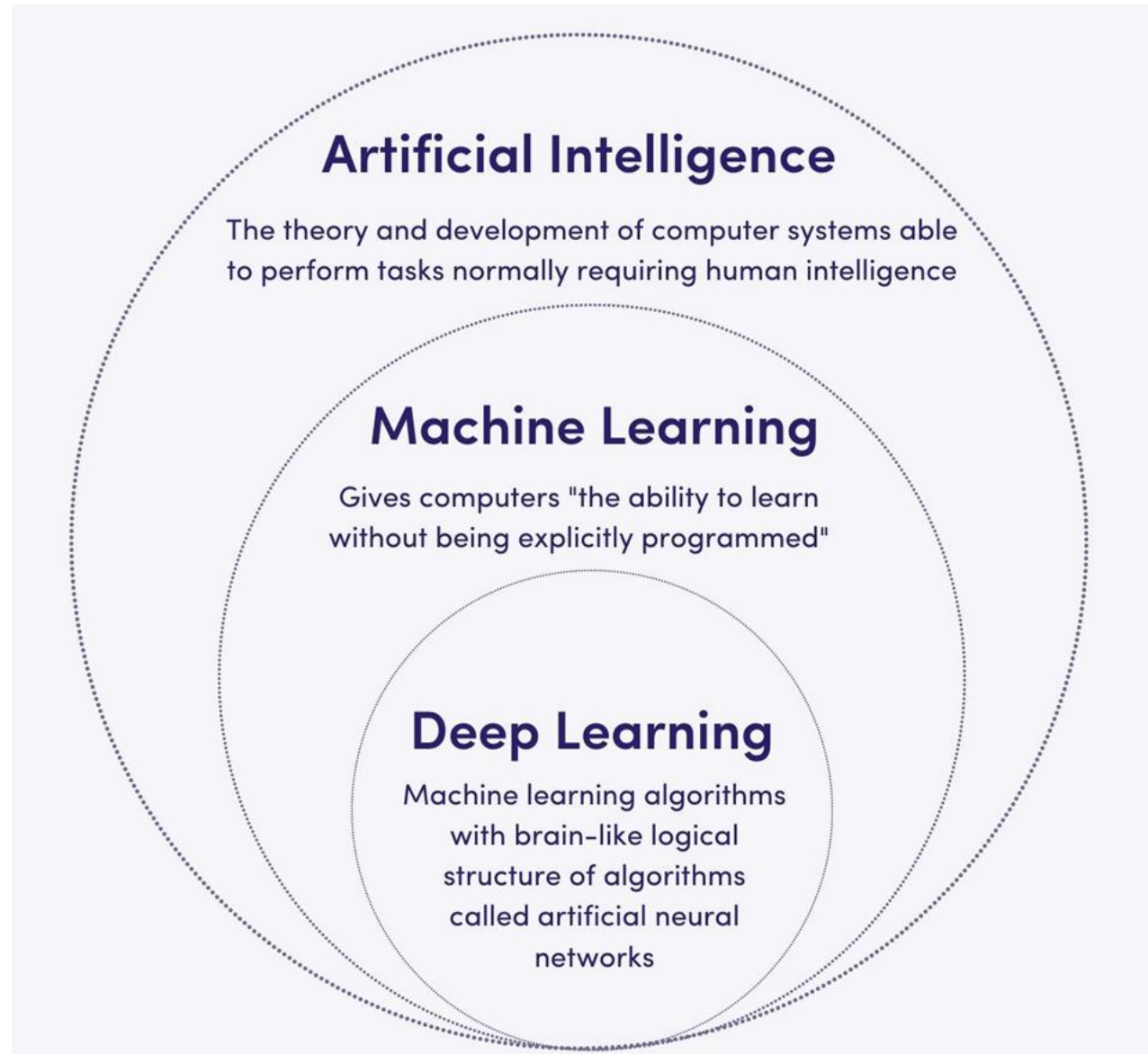
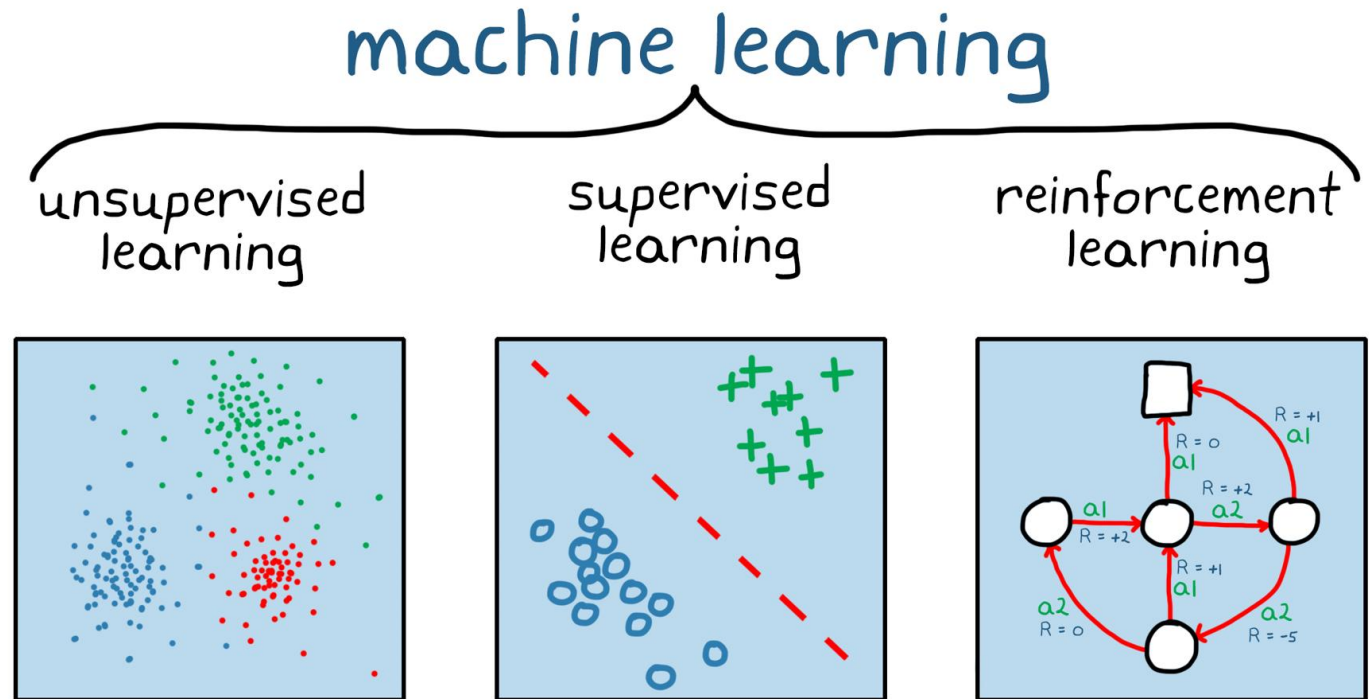


Strojno učenje (Machine Learning – ML)

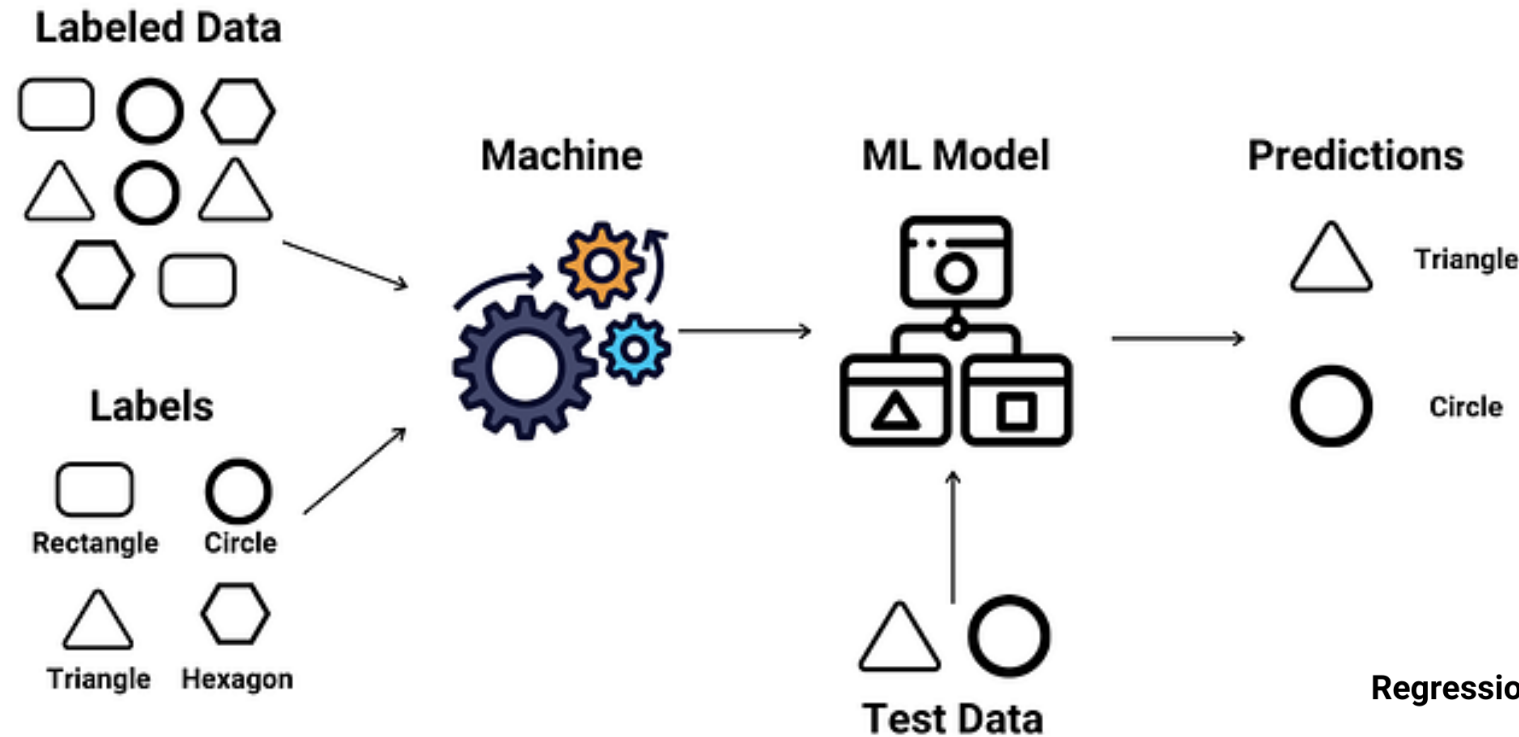


Delitev metod strojnega učenja

- Nadzorovano učenje (supervised learning)
 - Klasifikacija in regresija
- Nenadzorovano učenje (unsupervised learning)
 - Gručenje
- Spodbujevano učenje (reinforcement learning)

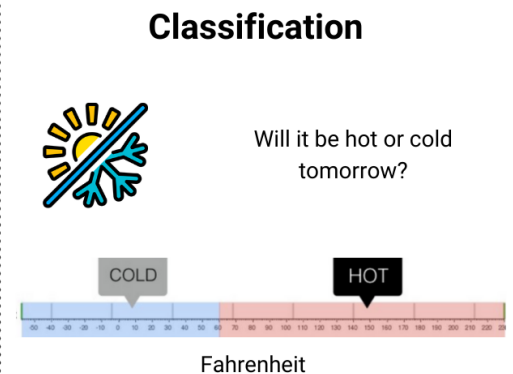
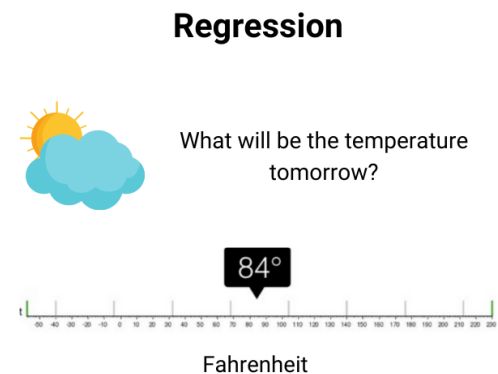


Delitev metod strojnega učenja



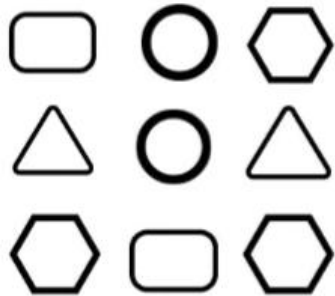
Classification where our result set consist of categories

Regression where the results are continuous values.



Delitev metod strojnega učenja

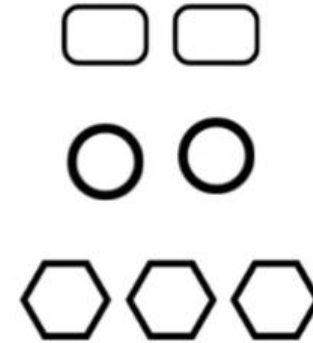
Unlabelled Data



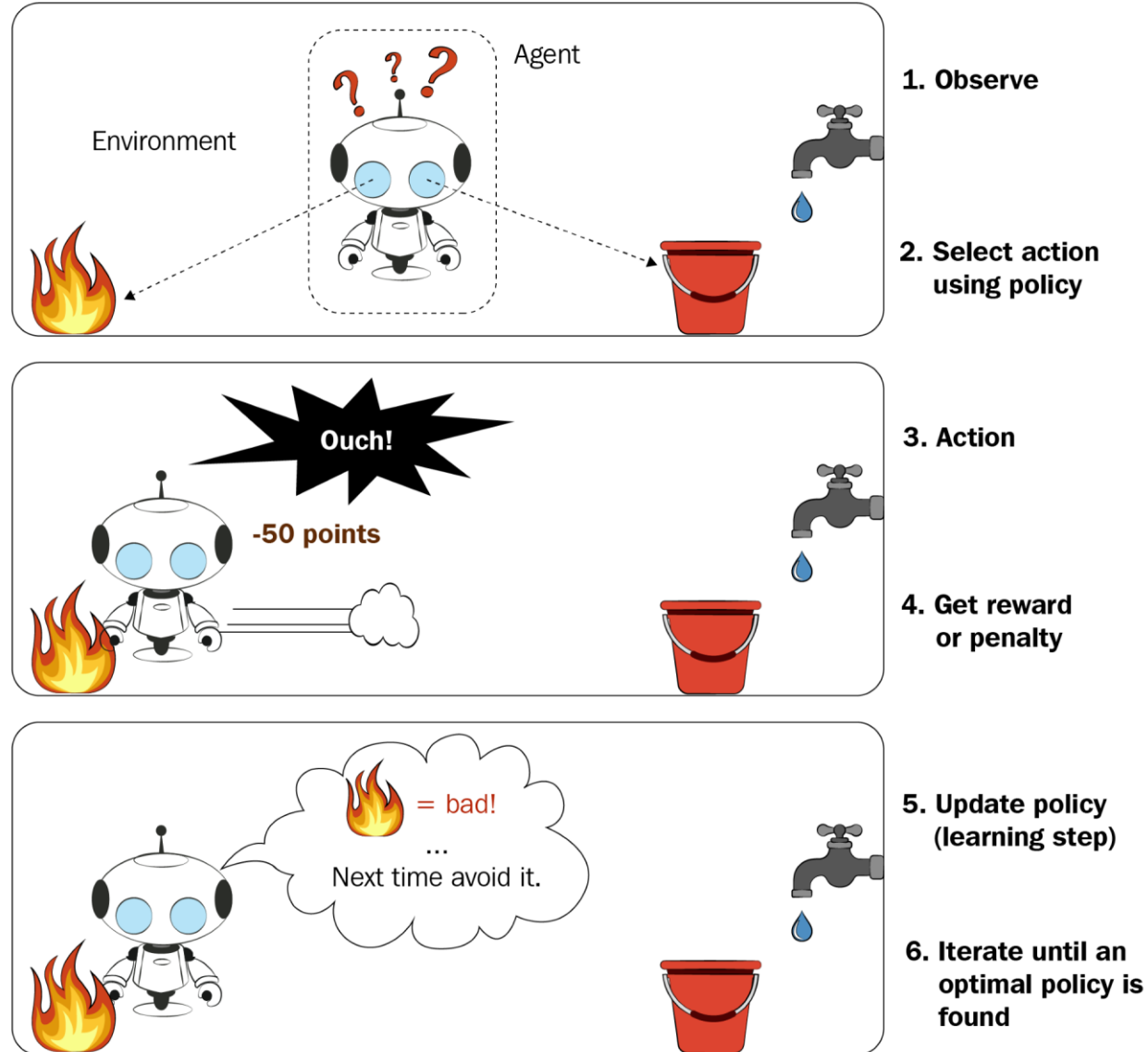
Machine



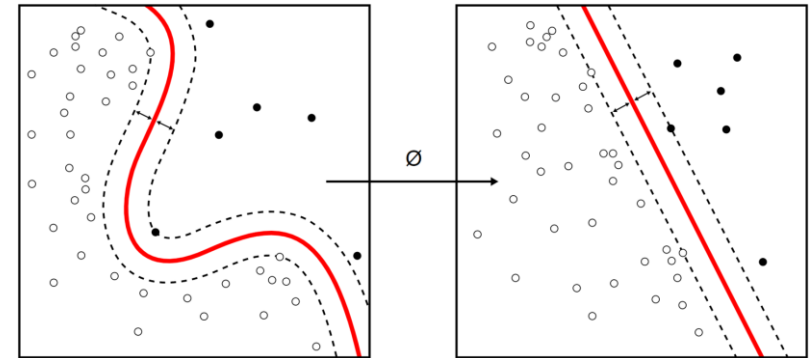
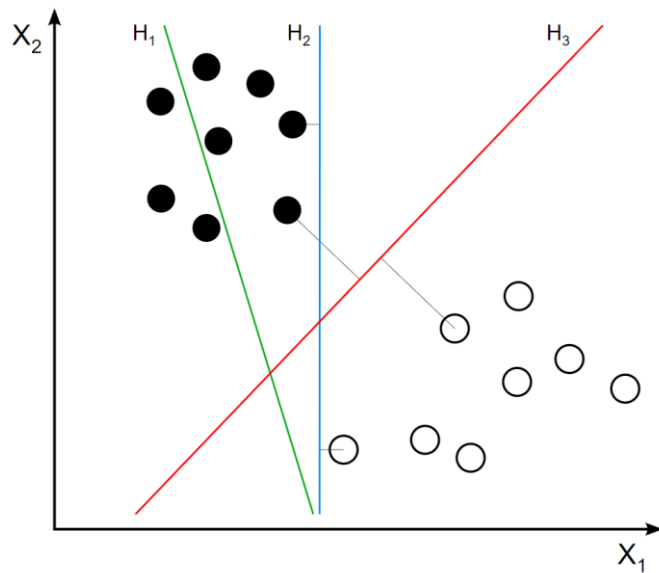
Results



Delitev metod strojnega učenja

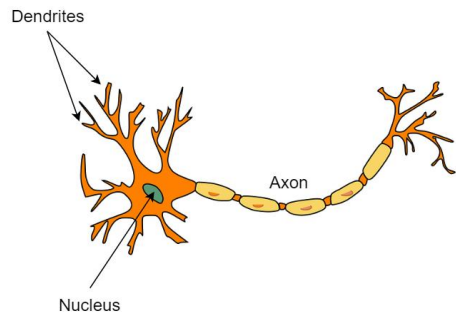


Metoda podpornih vektorjev - SVM

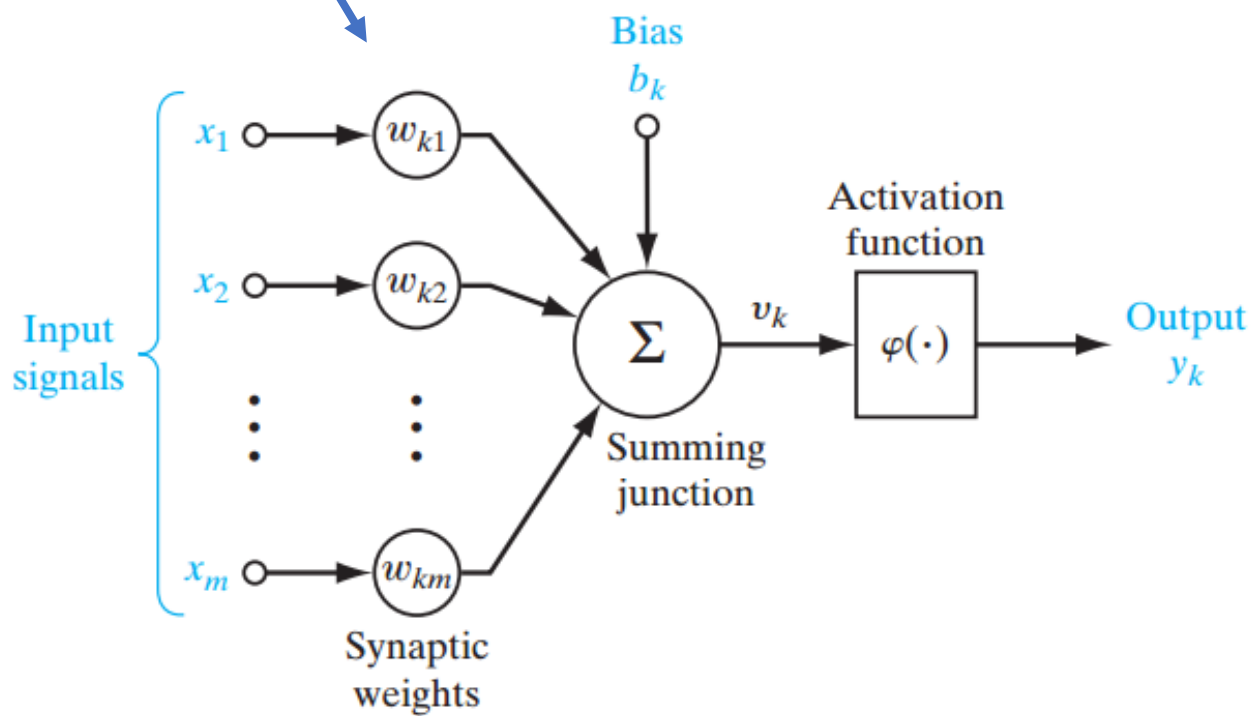


*kernel
trick*

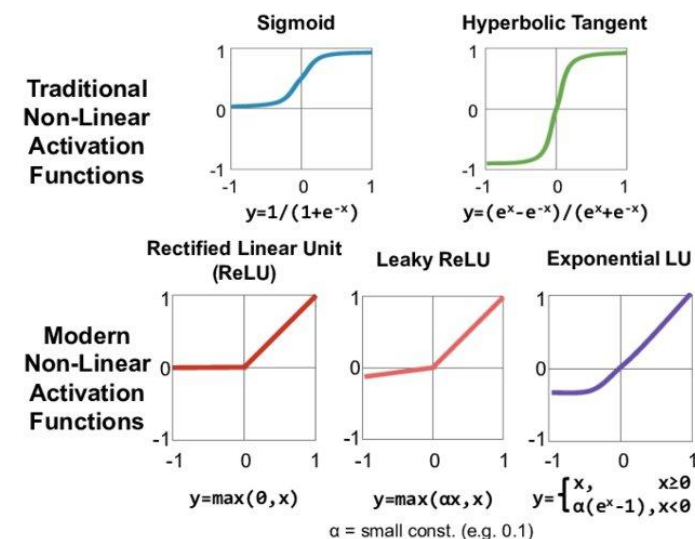
Perceptron – model umetnega nevrona



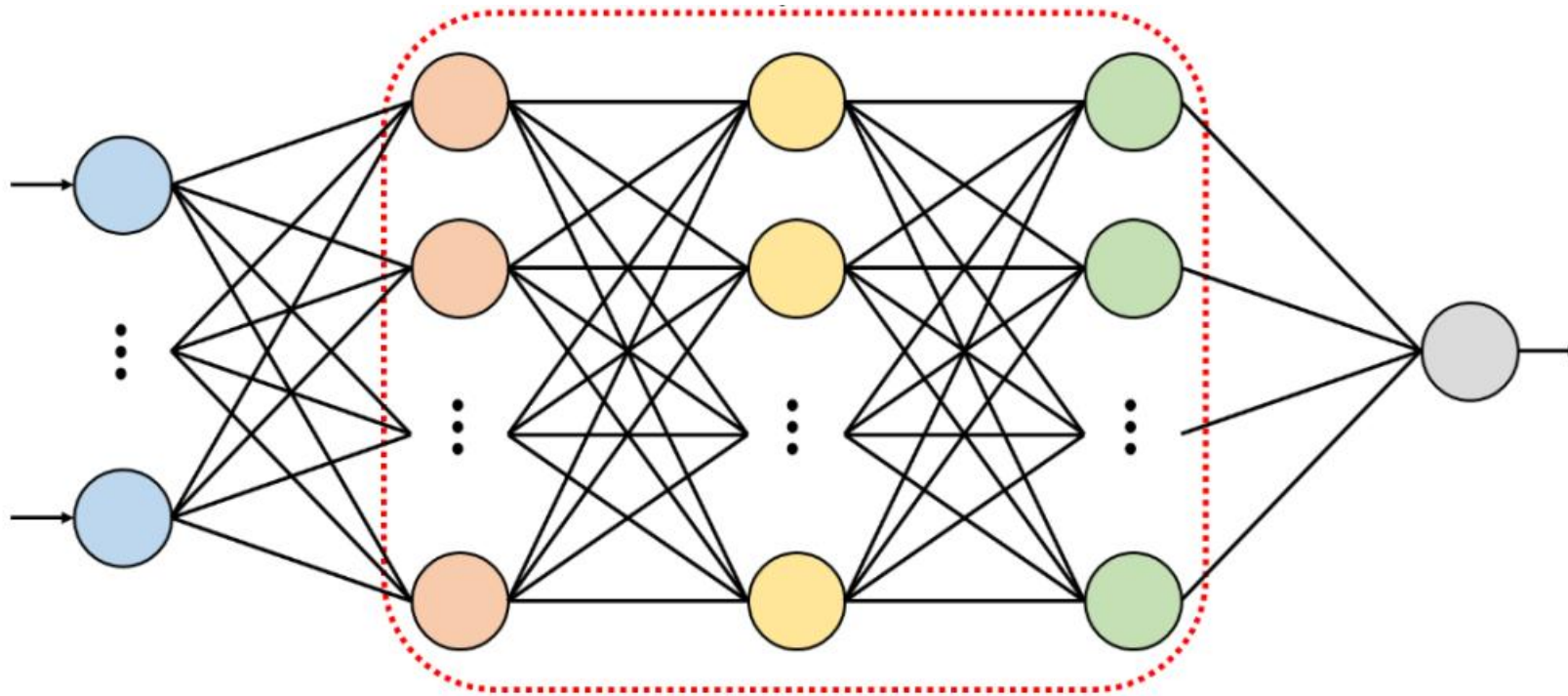
*Sinaptične uteži w_n hranijo
“znanje” perceptrona*



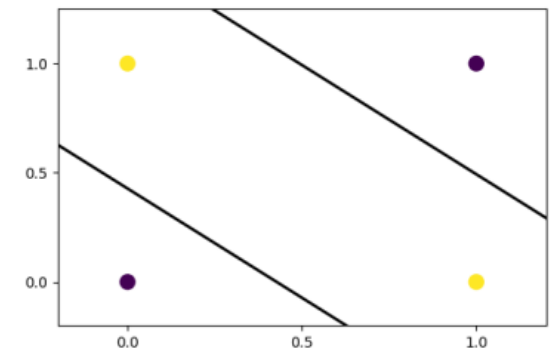
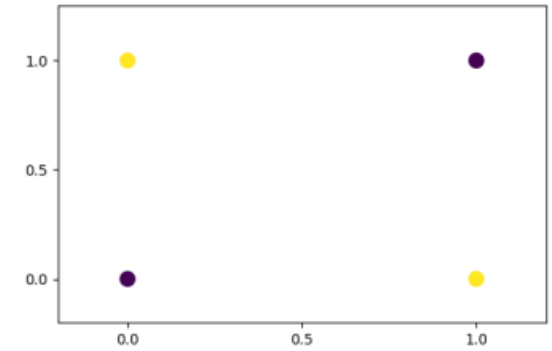
Aktivacijske funkcije



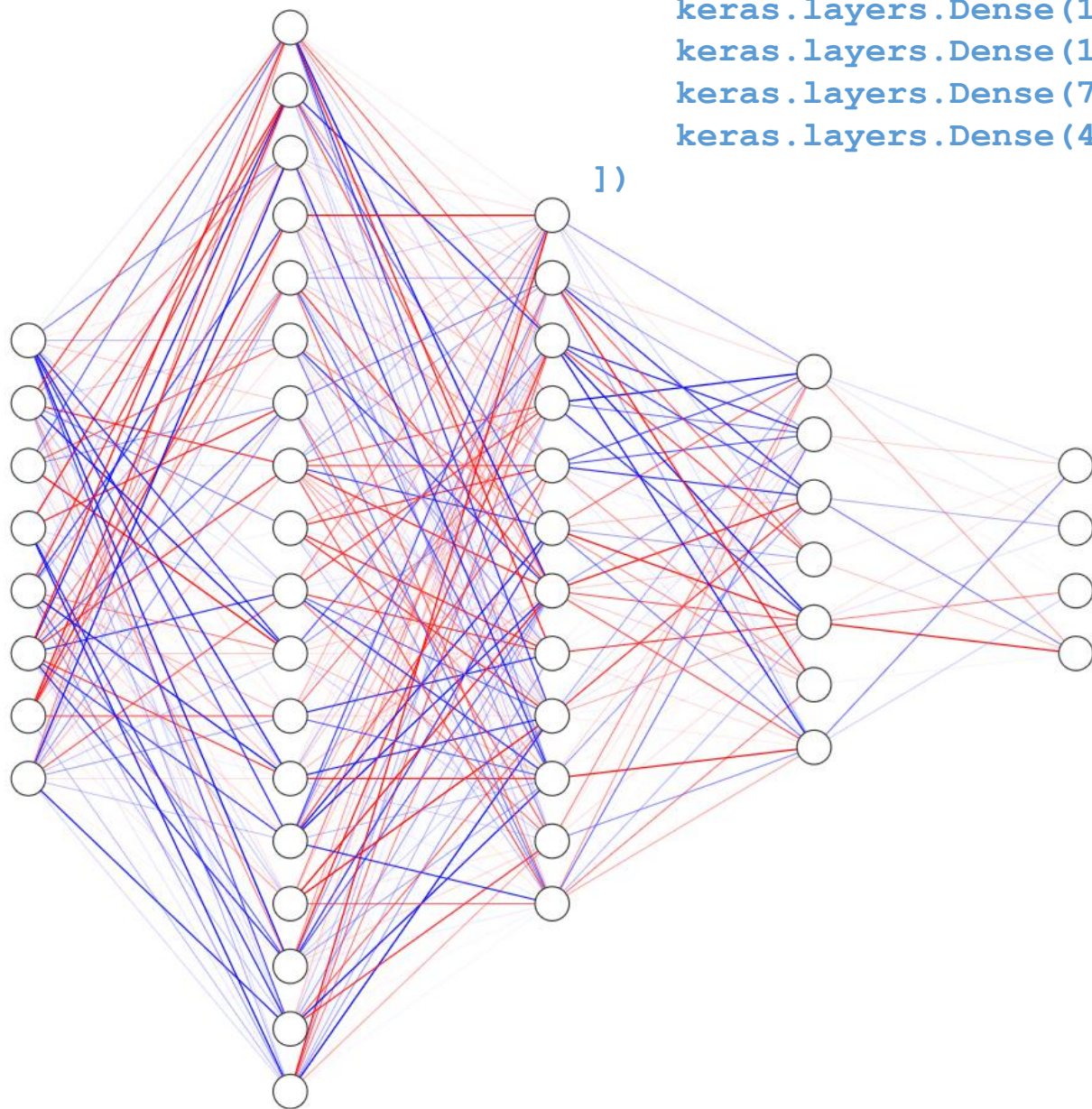
Večplastní perceptron



XOR



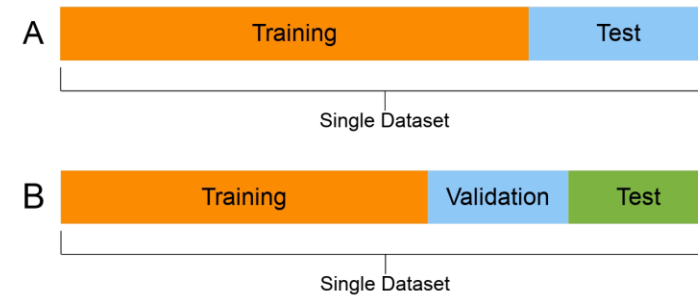

```
model = keras.Sequential([  
    keras.layers.Dense(18, activation = "relu", input_shape = (8,)),  
    keras.layers.Dense(12, activation = "relu"),  
    keras.layers.Dense(7, activation = "relu"),  
    keras.layers.Dense(4, activation = "sigmoid")  
])  
...
```



Umetna nevronska mreža
(neural network)
Model večplastnega perceptrona
(multi layer perceptron - MLP)

Postopek učenja

- Učna, validacijska in testna množica podatkov
 - Pari (vhodi ✓ - izhodi ???)
- Arhitektura nevronske mreže (plasti)
- Postopek učenja – optimizacija uteži $\mathbf{w}$
 - Gradientni sestop (SGD), ADAM, ...
- Križna validacija (k-fold cross validation)

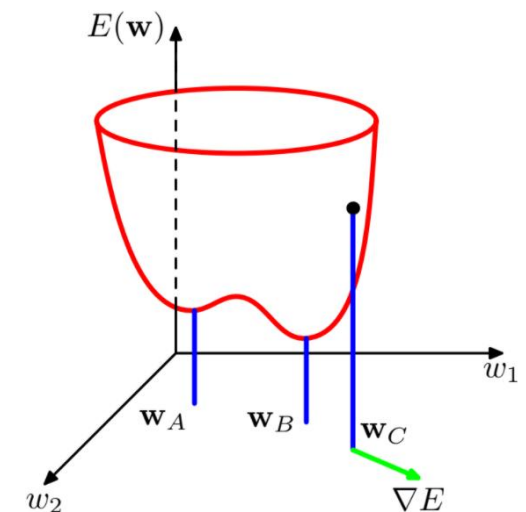
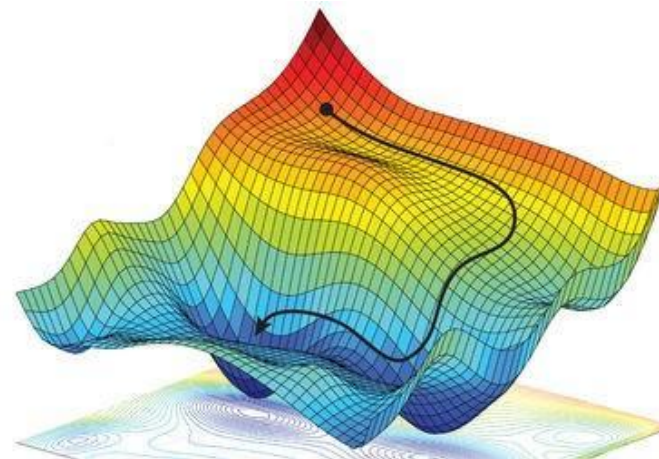
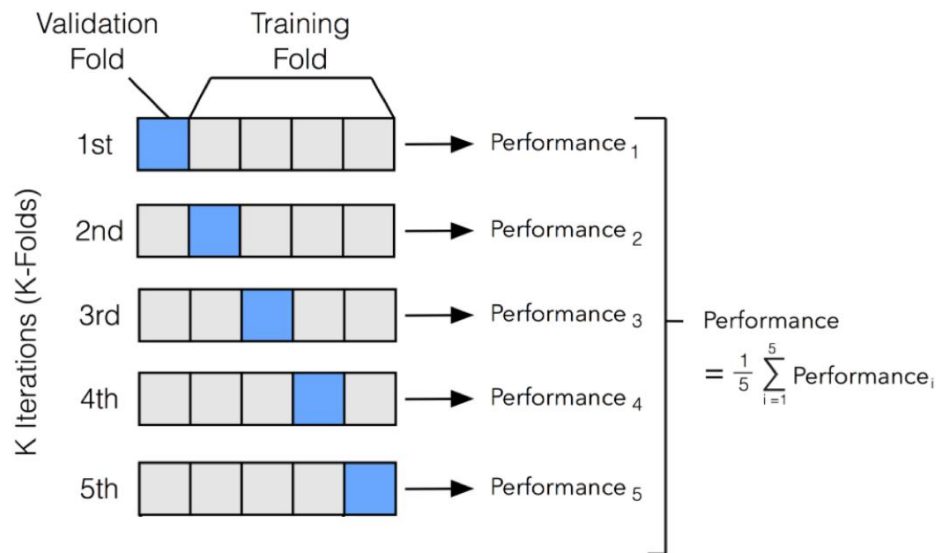


Numerical

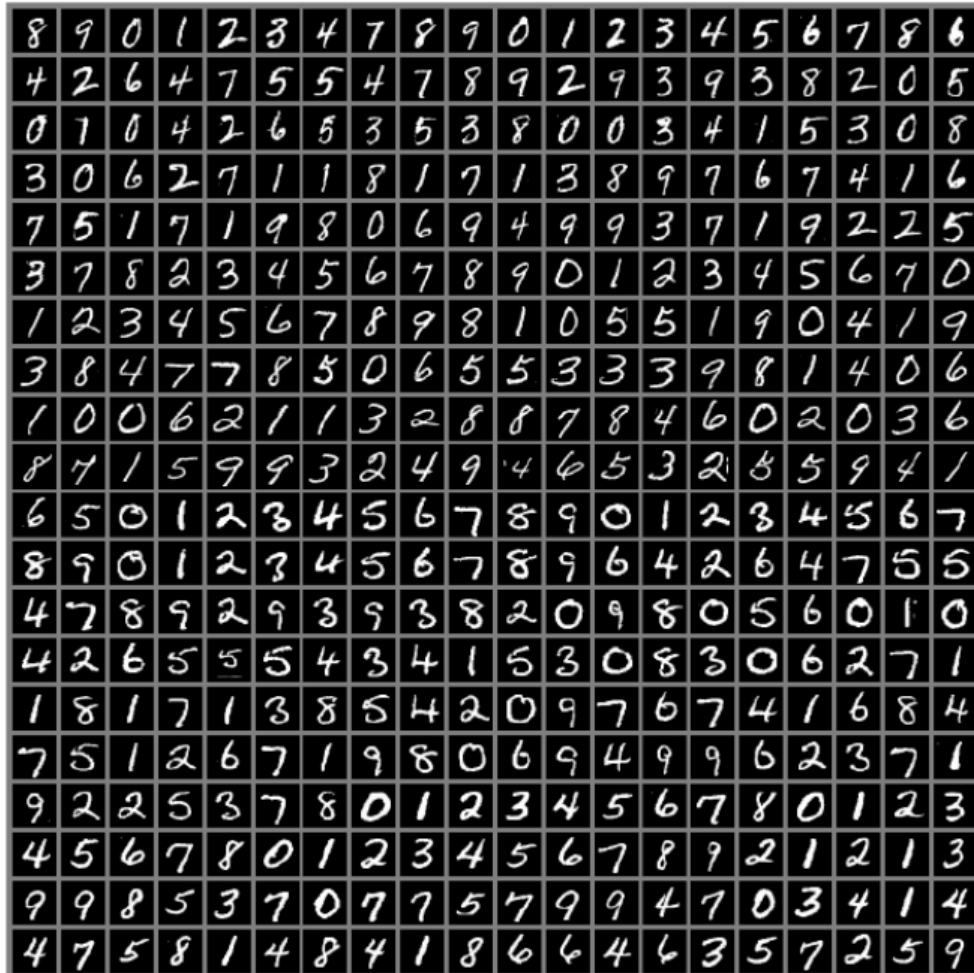
Word	ID
Red	0
Sticky	1
Clay	2

One-hot

Red	Sticky	Clay
1	0	0
0	1	0
0	0	1



Primer nabora podatkov za klasifikacijo - MNIST



slike

oznake

+ [8, 9, 0, 1, 2, 3, ...]

SparseCategoricalCrossentropy

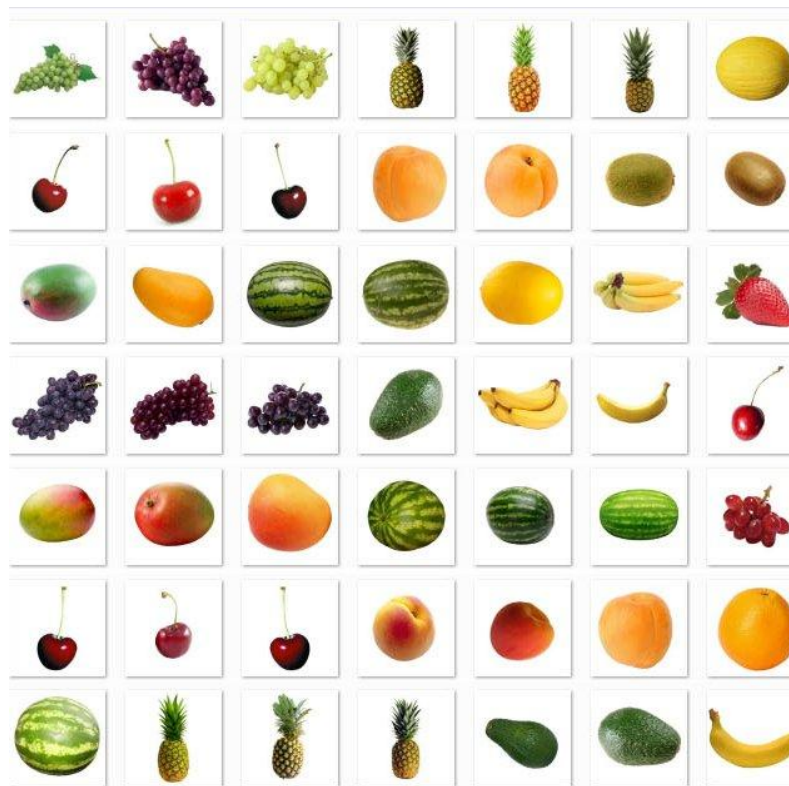
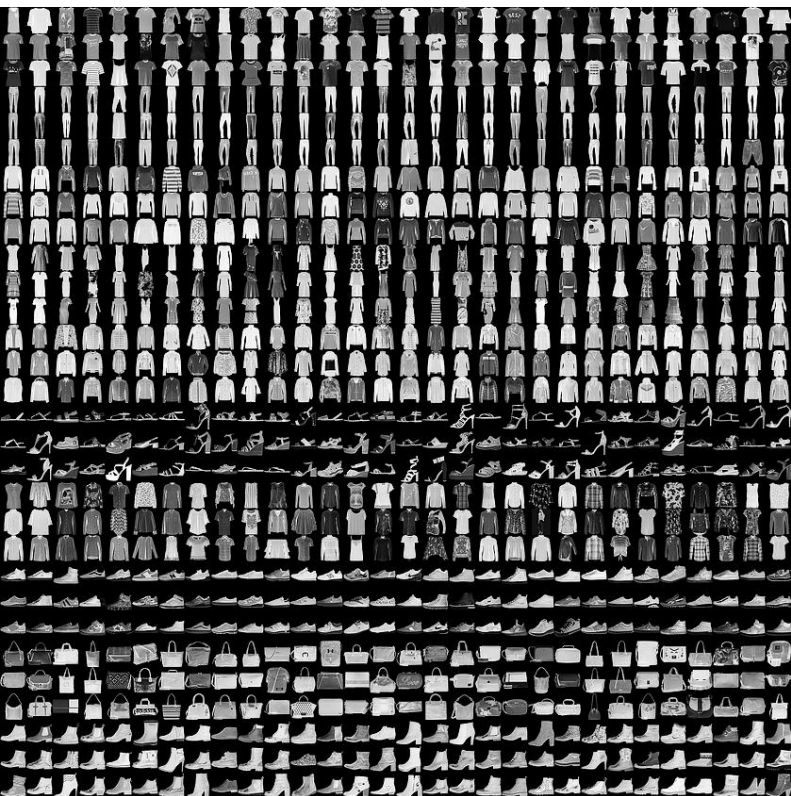
ali

CategoricalCrossentropy + OneHot pretvorba

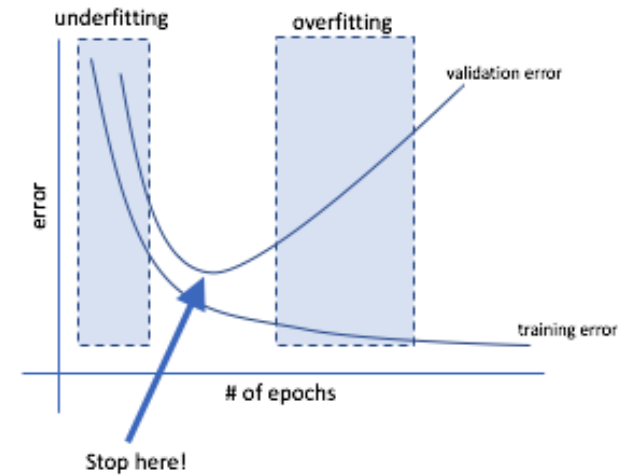
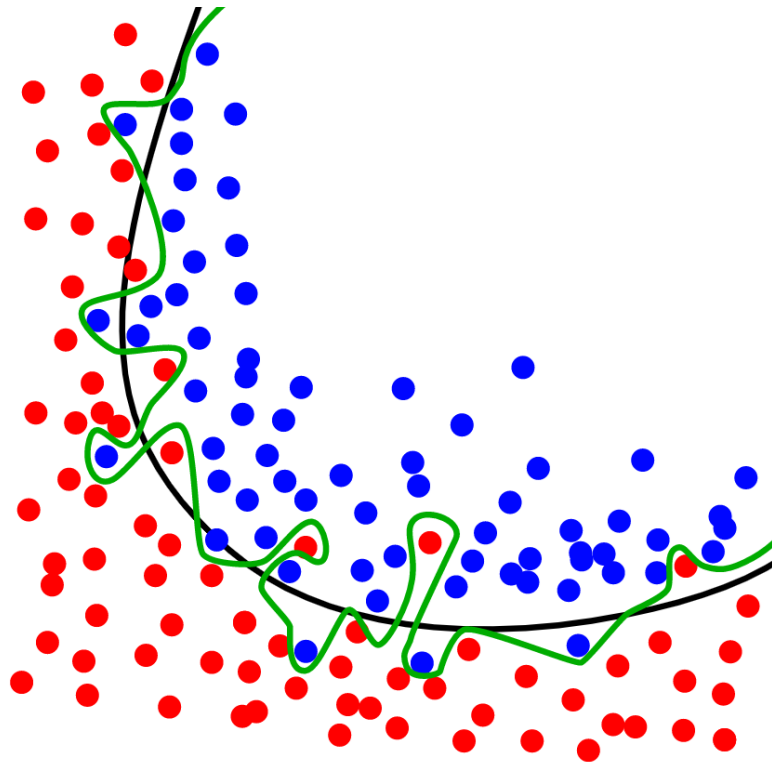
Softmax aktivacija na zadnjem sloju

ali

from_logits=true

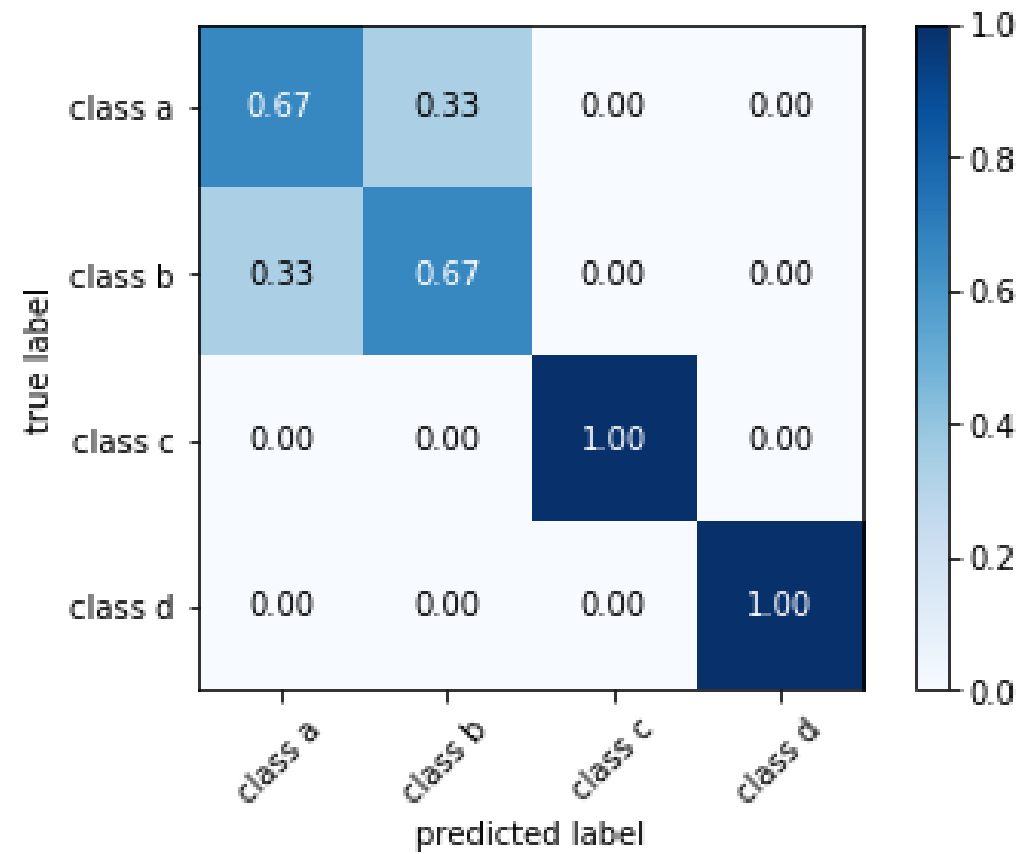


Prekomerno učenje - overfitting



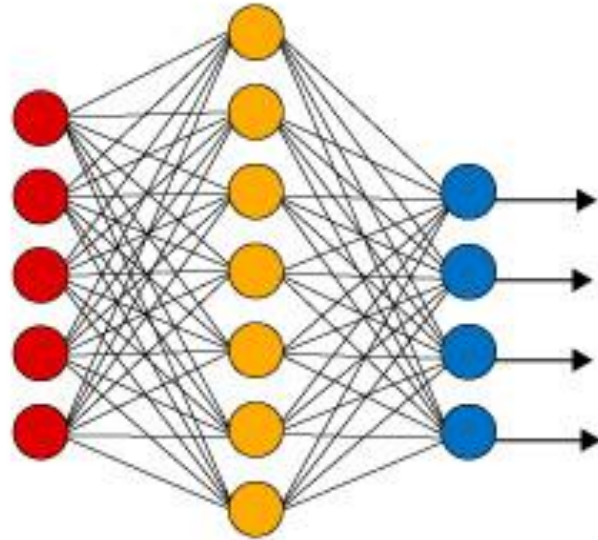
Zakaj je prenaučenosť nezaželjena?

Matrika zmede

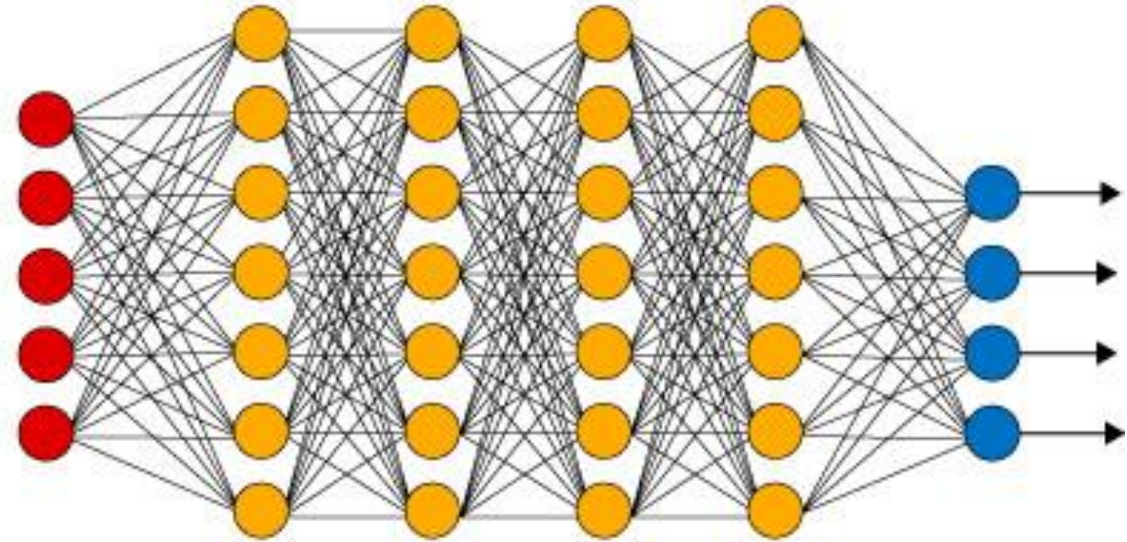


Globoke nevronske mreže - DNN

Simple Neural Network



Deep Learning Neural Network

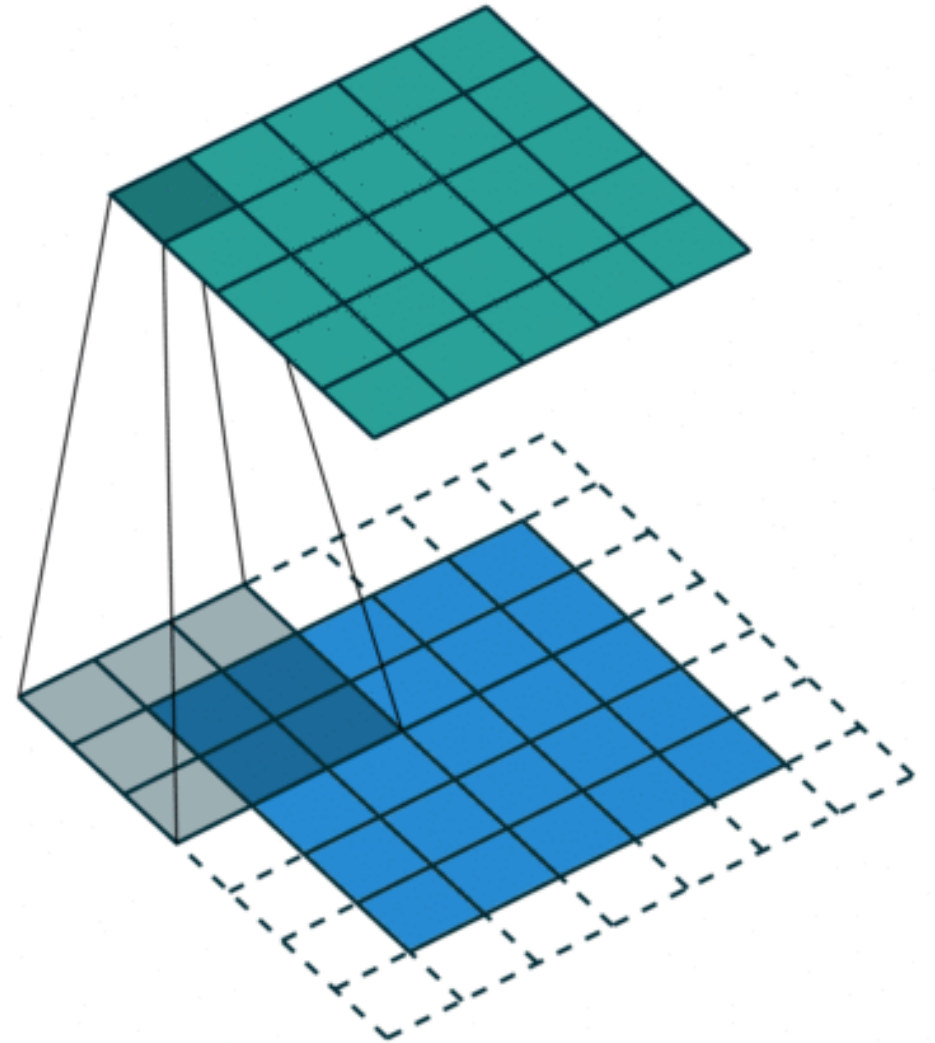
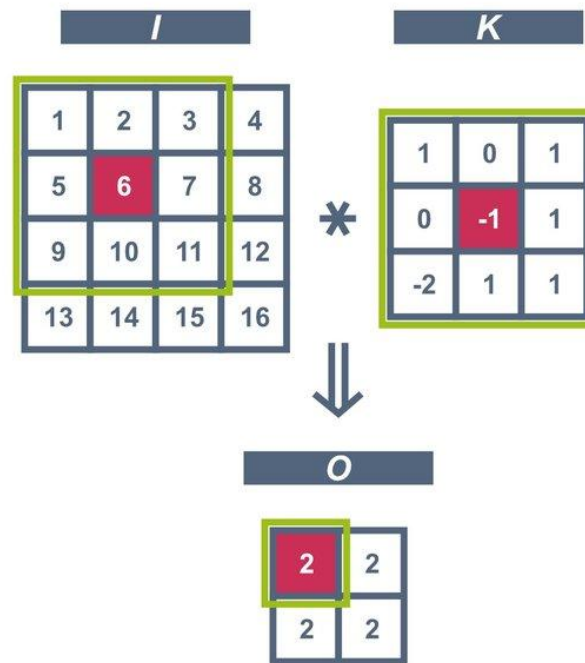


● Input Layer

● Hidden Layer

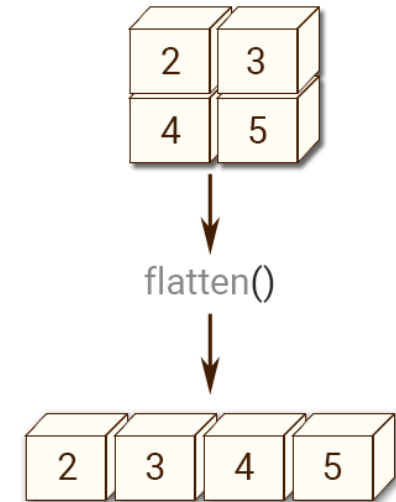
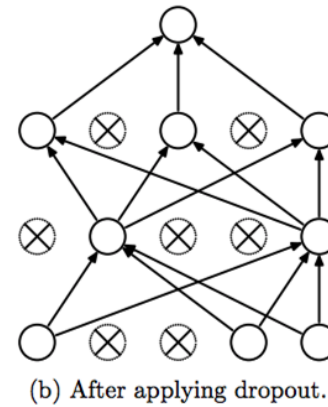
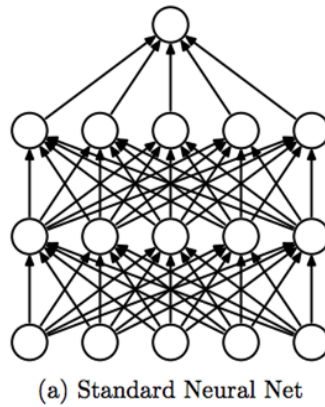
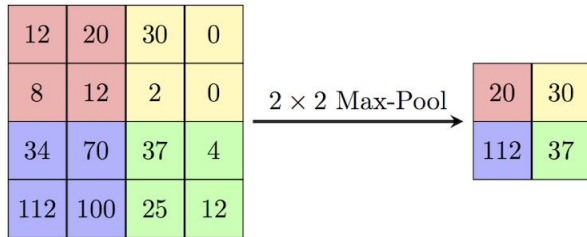
● Output Layer

Operacija konvolucije - convolution



Zakaj iz slike metulja uporabiti samo robove?

Sloji nevronske mreže



Konvolucijske nevronske mreže - CNN

